Requirements Specification Document

Student Academic Progress System

RUNTIME TERROR

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CMPS 4113: Software Engineering	Spring 2025	v1.1	Requirements Specification

Revision History

Version	Date	Author(s)	Description of Change
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v1.0	Feb 2025	Runtime Terror	First draft submitted
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Table of Contents

1. **Introduction / Overview**
   - 1.1 Purpose of Document
   - 1.2 Scope of Project
   - 1.2.1 Main Objective
   - 1.2.2 Specific Goals
   - 1.3 Overview of Document
2. **Users**
   - 2.1 Who Are Users
   - 2.2 Use Cases
   - 2.3 Scenarios
3. **System**
   - 3.1 Development Environment
   - 3.2 Target Environment
   - 3.3 Functional Requirements
   - 3.3.1 Issues
   - 3.3.2 Major Subsystems
   - 3.3.3 Major Functions
   - 3.3.4 Major Classes (OOA Class Diagram)
   - 3.3.5 Minor System Functions
   - 3.4 User Interface Specifications
   - 3.5 Non-Functional Requirements
   - 3.5.1 Management Constraints
   - 3.5.2 Technical Constraints
   - 3.5.3 Performance Requirements
   - 3.5.4 Security Requirements
   - 3.6 System Evolution / Maintenance
4. **Other Deliverables**
5. **Risks**

6. **Additional Considerations**
7. **Glossary**
8. **References**

1. Introduction / Overview

1.1 Purpose of Document

The purpose of this Requirements Specification Document is to clearly define the functional and non-functional requirements for the Student Academic Progress (SAP) System developed by Runtime Terror for CMPS 4113: Software Engineering. This document describes the SAP System and its expected behaviors so that all developers and project team members fully understand the system's purpose, capabilities, and expectations.

This document serves as the authoritative reference for what the system shall do, how it shall perform, and the constraints under which it must operate. It guides all development decisions, QA testing, and final evaluation throughout the project lifecycle.

1.2 Scope of Project

The Student Academic Progress (SAP) System is a secure web-based application designed to support academic advising and improve student success outcomes. The system shall allow students to monitor their academic progress toward degree completion, identify at-risk standing, and receive academic guidance through an interactive dashboard. The system focuses strictly on academic progress tracking and advising functions within a single academic institution and is not intended as a general purpose learning management system.

The SAP System will store and manage student academic records. It will interface with course and major requirement data to perform automated degree audits and generate actionable alerts and recommendations for students.

1.2.1 Main Objective

The main objective of the SAP System is to provide a centralized platform that tracks student academic progress, identifies potential academic risks, and assists students in successfully completing their degree requirements. The system consolidates course history, GPA calculations, credit tracking, and degree audit functions into a single accessible application.

1.2.2 Specific Goals

The specific goals of the SAP System shall include the following:

- Store and display complete academic history
- Track completed and in progress courses with grades and credit hours
- Calculate and display the current cumulative GPA
- Track total earned credit hours against degree requirements
- Define and store degree requirements organized by major
- Perform automated degree audits comparing student records to major requirements
- Display degree completion percentage as a visual progress indicator
- Identify missing or incomplete degree requirements
- Recommend future courses based on remaining requirements and completed prerequisites

1.3 Overview of Document

This document describes the users, system requirements, technical environment, risks, and deliverables associated with the Student Academic Progress System. Section 2 defines the primary users, their use cases, and example interaction scenarios. Section 3 covers the complete system specification including environments, functional requirements, subsystems, OOA class diagram, and non-functional requirements. Sections 4 through 6 address deliverables, risks, and future considerations. Sections 7 and 8 provide the glossary and references.

2. Users

2.1 Who Are Users

The primary users of the Student Academic Progress System are students enrolled at the academic institution. Students use the system to track academic progress, review degree completion status, plan semester course schedules, and enter course completions with associated grades. The system is designed to give students a clear, centralized view of their academic standing without needing to contact an advisor for basic progress information.In future iterations, academic advisors may be granted system access to review student records, run degree audits, and identify at risk students. T

2.2 Use Cases

The following use cases define the primary interactions a student will have with the SAP System. The Use Case Diagram immediately below illustrates the full set of use cases and the relationships between them, including include and extend dependencies.

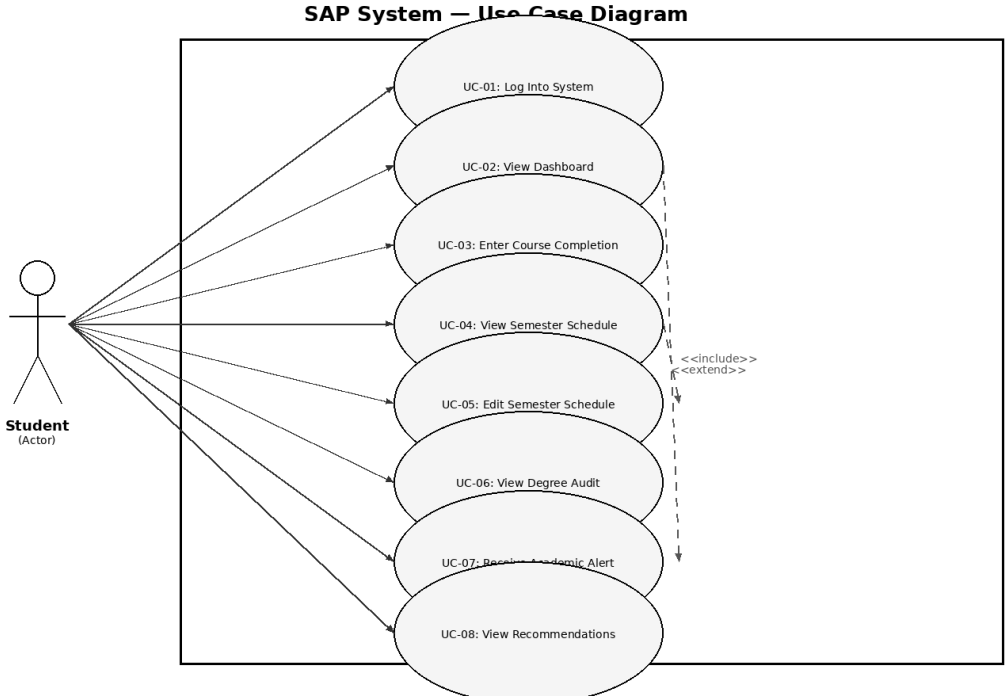

Figure 1 — SAP System Use Case Diagram

The table below describes each use case in detail:

Use Case ID	Use Case Name	Actor	Description
UC-01	Log Into System	Student	Student authenticates with credentials and gains access to their personal dashboard.
UC-02	View Dashboard	Student	Student views GPA, total earned credits, degree completion percentage, and active alerts.
UC-03	Enter Course Completion	Student	Student enters completed course data including course name, credit hours, and grade received.
UC-04	View Semester Schedule	Student	Student views current and past semester courses organized by term.
UC-05	Edit Semester Schedule	Student	Students add or remove courses from their planned schedule. from
UC-06	View Degree Audit	Student	Student views audit results showing completed and remaining requirements with completion percentage.

Use Case ID	Use Case Name	Actor	Description
UC-07	Receive Academic Alert	Student	System generates and displays an alert for student when GPA falls below threshold or credit progress is insufficient.
UC-08	View Recommendations	Student	Student views recommended future courses filtered by remaining requirements and completed prerequisites.

2.3 Scenarios

The following scenarios illustrate how real students will interact with the SAP System during typical sessions. These scenarios validate that the system's functional requirements address genuine student needs.

Scenario 1: Student Reviews Dashboard and Receives Alert

A junior Computer Science student logs into the SAP System after the midterm grading period. The dashboard immediately displays their current cumulative GPA (2.8), total earned credits (72 of 128), and a red alert banner indicating the GPA has fallen below the 3.0 major requirement threshold. The student navigates to the degree audit panel, which identifies two unfulfilled core requirements. The system then recommends CMPS 3305 and CMPS 4113 as the next courses based on the student's completed prerequisites. The student now has a clear understanding of their academic standing and a defined action plan.

Scenario 2: Student Enters Completed Courses

A sophomore logs in after completing the Fall semester and navigates to the course entry section. The student enters three courses, providing the course name, credit hours, and final grade for each. The system automatically recalculates the cumulative GPA and updates the degree completion percentage. The degree audit now marks those courses as fulfilled requirements and removes them from the course recommendation list. The student's record is fully updated without advisor involvement.

Scenario 3: Student Plans Upcoming Semester Schedule

A freshman logs in during the course planning period for the upcoming semester. The student navigates to the Semester Schedule section and reviews the system's course recommendations, which are filtered to show only courses whose prerequisites have already been satisfied. The student selects courses from the recommendation list and adds them to the planned schedule for the upcoming term. The dashboard updates to reflect projected credit hours for the new semester.

3. System

3.1 Development Environment

The SAP System will be developed using modern web technologies selected to support rapid Agile prototyping while maintaining code quality, version control, and collaborative development across the team. It is designed for desktop use on PC and Mac. It supports Windows 10+, macOS 10.15+, and Ubuntu 20.04+, is not designed for phones or tablets in this version, and can be accessed through Chrome, Firefox, Edge, or Safari. All data stays local, no cloud, no external systems

Component	Technology / Tool	Purpose
Version Control	GitHub	Source code management and team collaboration
Code Editor / IDE	Visual Studio 2026, VS Code	Primary development environment for all team members
Backend Language	C++, C#, Python, .Net Framework	Business logic, GPA calculations, degree audit processing
Database	SQLite / MySQL (TBD)	Relational storage for students, courses, requirements
Frontend	HTML, CSS, JavaScript, and maybe figma tool	User interface, dashboards, and form interactions
Testing	PyTest / Manual QA	Unit testing and quality assurance validation
Hardware	Team member laptops + shared device	Local development, testing, and final presentation

3.2 Target Environment

The SAP System will run as a locally hosted web application and will be accessed through a standard web browser on any modern desktop operating system (Windows, macOS, Linux). No software installation is required on the student's device beyond a compatible browser. Mobile browser access is not a primary target for v1.0 but is planned for future evolution. All student data is stored and managed locally with no dependency on external institutional systems in this release.

3.3 Functional Requirements

The functional requirements define all behaviors and capabilities the SAP System must provide. Each requirement is specific and testable, derived directly from the system scope, use cases, and scenarios defined above.

3.3.1 Issues

The following known challenges were identified during requirements analysis and must be addressed during development:

- **Data Accuracy:** Student records entered manually must pass input validation to prevent incorrect grades or credit hours from corrupting GPA calculations, degree audit results, and alert generation
- **GPA Calculation Consistency:** The GPA formula must be implemented correctly and uniformly across all use cases. The QA team will validate GPA accuracy using test cases with known expected results as a primary acceptance criterion
- **Degree Audit Complexity:** Mapping completed coursework to degree requirements across categories and prerequisites is inherently complex. The initial version targets Computer Science degree requirements with extensibility planned for other majors
- **Future Integration Challenges:** Integration with institutional registration systems in later versions will require careful handling of data format compatibility and authentication protocols

3.3.2 Major Subsystems

The SAP System is composed of five major subsystems, each responsible for a distinct area of functionality that communicates through the backend to deliver a unified user experience.

Subsystem	Description	Key Responsibilities
Academic Records Management	Handles storage, retrieval, and display of all student academic data.	Course entry, grade storage, credit hour tracking, history display
Degree Audit	Compares completed coursework against degree requirements.	Requirement mapping, completion percentage, missing requirement identification
Alert and Notification	Monitors GPA and credit progress; generates alerts on threshold violations.	GPA monitoring, credit alert generation, dashboard warning indicators
Dashboard Reporting	Aggregates and displays all academic data in a unified student view.	GPA display, progress bar, course lists, alert banners, recommendations

3.3.3 Major Functions

The following major functions represent the core capabilities the SAP System must deliver. Each function is a critical, testable behavior tied directly to a use case in Section 2.

Function ID	Function Name	Description
F-01	Display Academic History	Retrieve and display all completed and in-progress courses including grades and credit hours.
F-02	Calculate GPA	Compute cumulative GPA: sum(grade points × credit hours) ÷ total credit hours attempted.
F-03	Track Earned Credits	Accumulate total earned credit hours from passed courses and compare against the degree credit requirement.
F-04	Perform Degree Audit	Compare completed coursework against declared major requirements and produce a fulfillment report.
F-05	Display Completion Percentage	Calculate and display degree completion percentage as a visual progress bar on the dashboard.
F-06	Identify Missing Requirements	List all degree requirements not yet fulfilled by the student's completed coursework.
F-07	Recommend Future Courses	Recommend courses satisfying remaining requirements, filtered by completed prerequisites.
F-08	Generate Academic Alerts	Generate and display alerts when cumulative GPA falls below 2.0 or credit hour progress is insufficient.
F-09	Display Dashboard	Present a unified view aggregating GPA, earned credits, degree completion, alerts, and course recommendations.

3.3.4 Major Classes (OOA Class Diagram)

The following five major classes form the object-oriented analysis (OOA) data model of the SAP System. Each class encapsulates a key domain entity with defined attributes and relationships that support all system functions. The OOA Class Diagram below illustrates these classes, their attributes, key methods, and the associations between them including multiplicity.

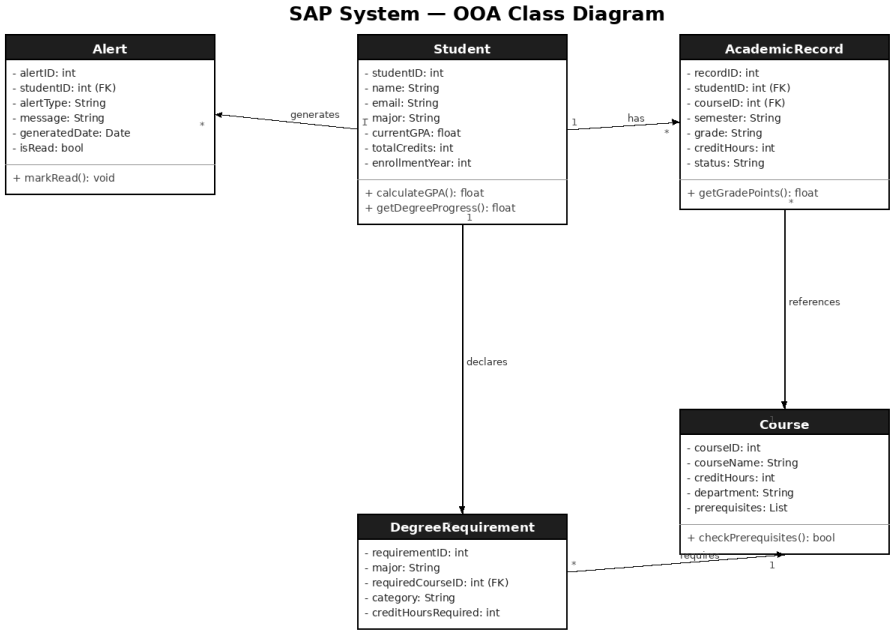

Figure 2 — SAP System OOA Class Diagram

The five major classes and their roles within the system are summarized below:

- **Student:** The central entity. Stores personal and academic identity information and owns all AcademicRecord and Alert instances for that student. Provides calculateGPA() and getDegreeProgress() operations
- **AcademicRecord:** Represents a single course enrollment or completion for a student. Links the Student to a Course with a grade, credit hours, semester, and completion status. Used by the degree audit and GPA calculation functions
- **Course:** Represents a course offered by the institution. Stores the credit hours, department, and prerequisite course list. Referenced by both AcademicRecord and DegreeRequirement.
- **DegreeRequirement:** Defines a required course within a specific major and category. Compared against AcademicRecord entries during the automated degree audit to determine fulfillment status
- **Alert:** Stores a warning generated by the system when a student's GPA or credit progress falls below defined thresholds. Belongs to a Student and is displayed on the Dashboard

3.3.5 Minor System Functions

In addition to the major functions, the SAP System shall support the following minor functions that contribute to correct operation, maintainability, and future extensibility:

- **Update Course Records:** Students may update previously entered course records subject to input validation rules before degree audit finalization

- **Update Degree Requirements:** Administrators (in future versions) may update the degree requirements table when curriculum changes occur, keeping the degree audit accurate over time.
- **Display Dashboard Statistics:** The dashboard shall display aggregate statistics including total courses completed and projected graduation term, based on current credit pace
- **Determine Course Prerequisites:** The system shall evaluate prerequisite completion before recommending any course to the student, automatically filtering out ineligible options

3.4 User Interface Specifications

The user interface shall prioritize simplicity, readability, and accessibility. All interactions are browser based, requiring no installation on the student's device. The color scheme will be professional and neutral. The following UI components are required in the initial release:

UI Component	Description
Student Dashboard	Primary view after login. Displays GPA, degree completion progress bar, earned credit total, and active alert indicators in a card-based layout.
GPA Display	Prominent numeric display of cumulative GPA with color coding: green (3.0+), yellow (2.0–2.99), red (below 2.0).
Degree Progress Bar	Horizontal bar showing degree completion percentage with a numeric label alongside.
Alert Banner	Red warning banner at the top of the dashboard when GPA or credit thresholds are not met.
Course History Table	Table listing all courses with columns for name, credit hours, grade, semester, and completion status.
Degree Audit Panel	Two-column layout showing fulfilled requirements (checkmark) and unfulfilled requirements (X) grouped by category.
Course Recommendation List	Ordered list of recommended future courses showing course name, credit hours, and prerequisite status.
Semester Schedule View	List view of current and planned semester courses with add and remove functionality.

3.5 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints the SAP System must satisfy independently of any specific function.

3.5.1 Management Constraints

- The system must be developed within the Spring 2025 academic project timeline as defined in the Asana task board

- The project shall follow Agile sprint methodology with weekly sprint reviews and a Kanban style task board tracking To Do, In Progress, Testing, and Completed states
- All team members shall participate in code reviews before any feature is merged into the main branch
- Scope changes must be reviewed and approved by the Project Leader before any implementation begins

3.5.2 Technical Constraints

- The system must operate as a locally hosted web application using a relational database and must support all major standard web browsers (Chrome, Firefox, Edge, Safari)
- The system shall not depend on external APIs, cloud services, or third-party data sources in v1.0
- All source code shall be managed on GitHub with a consistent branching strategy and meaningful commit messages

3.5.3 Performance Requirements

- All standard user requests, page loads, GPA calculation, and degree audit display shall respond within five seconds under normal operating conditions
- GPA calculations shall be completed on demand with no perceptible delay to the user

3.5.4 Security Requirements

- On their own devices, which they secure

3.6 System Evolution / Maintenance

While the current version is a standalone, locally hosted application, it has been designed with extensibility in mind. Planned future enhancements include integration with institutional registration systems and academic databases, expansion to additional academic majors beyond Computer Science, mobile application development for iOS and Android, and an academic advisor portal with read only access to assigned student records and degree audit results.

4. Other Deliverables

In addition to the working SAP System software, the following documents and artifacts will be produced as part of this project:

Deliverable	Description	Responsible
Project Plan / Outline	Defines project scope, schedule, team structure, risk management, and tracking mechanics.	All team members
Requirements Specification	This document. Defines all functional and non-functional requirements for the SAP System.	All team members
System Design Documentation	Architecture diagrams, database schema, and class diagrams showing the full system structure.	Backend + Database
Testing Plan / Report	Test cases for all major functions; document QA testing results.	QA Tester
User Manual	End user guide explaining how to use the dashboard, enter courses, and read degree audit results.	Frontend + QA
Project Presentation	Final demonstration of the working system to the class and Stringfellow.	All team members

6. Additional Considerations

The SAP System has been designed with scalability and usability in mind to support future enhancements and a growing user base. The following considerations should guide design decisions that make the system easier to extend in future development cycles:

- **Scalability:** The system architecture shall support adding additional degree plans for other majors without requiring significant database restructuring. Degree requirements shall be stored as data records rather than hard-coded logic so that new majors can be added by inserting rows rather than changing code
- **Mobile Readiness:** While mobile access is not required in v1.0, the frontend should apply responsive design principles so the system can be more easily adapted for mobile browsers or wrapped in a native mobile application in future releases.
- **Multi-User Security:** As the number of concurrent users grows in future versions, session management and access control mechanisms should be reviewed. Multi-factor authentication may be introduced at that stage.
- **Maintainability:** All code shall be well commented and organized into clear modules corresponding to the five major subsystems, simplifying future maintenance and developer onboarding.

7. Glossary

The following terms are used throughout this document and throughout the development of the SAP System:

Term	Definition
Academic Progress	Measurement of how much coursework a student has completed toward their degree.
Academic Record	A record of a student's enrolled or completed course including course name, credit hours, grade, and semester.
Agile Methodology	A project management approach using short development cycles (sprints) to allow flexibility and iterative improvement.
Alert System	A feature that notifies students when performance falls below defined thresholds, such as low GPA or insufficient credits.
Backend	The server-side portion of the application is responsible for processing data, running calculations, and managing system logic.
Credit Hours	Units represent the amount of academic coursework completed toward a degree. Each course carries a defined credit hour value.
Dashboard/GUI	The main user interface screen displays academic progress, GPA, credit totals, degree completion percentage, and alerts.
Database	A structured collection of stored data (students, courses, grades, requirements) that supports all system functions.
Degree Audit	An automated process comparing completed coursework against degree requirements to determine graduation progress.
DegreeRequirement	A defined course or set of courses that must be completed as part of a student's declared major program.
Frontend	The visual and interactive portion of the web application that users see and interact with through their browser.
GPA (Grade Point Average)	Numerical representation of academic performance: total grade points earned divided by total credit hours attempted.
Milestone	A significant checkpoint in the project timeline marking completion of a major deliverable or phase.
Prerequisite	A course that must be completed before a student may enroll in or be recommended for a subsequent course.
Role-Based Access Control	A security method restricting system access based on the user's assigned role (student, advisor, administrator).
SAP System	Student Academic Progress System the web-based application developed by Runtime Terror.
Scope	The defined boundaries of what the project will and will not include.

Term	Definition
Semester Schedule	A list of courses a student is enrolled in or plans to take during a given semester, stored within the SAP System.
Specification Document	A detailed statement outlining the software's functionality and its operational constraints.
Sprint	A short, time-boxed development cycle used in Agile methodology to complete a defined set of tasks.
SQL	Structured Query Language the standard language used to create, query, and manipulate relational databases.
System Integration	The process of combining frontend, backend, and database components so they function as one unified system.
Team Leader	Leader of the software development group responsible for scope, timeline, and team coordination.
Timeline	A schedule detailing a chronological sequence of project events and milestones.
User Testing	The process of having end users test the system to verify that usability and functionality meet defined requirements.

8. References

The following references were consulted during the development of this Requirements Specification Document:

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End of Requirements Specification Document — Runtime Terror — Spring 2025